

sage is received, its message counter is checked against the history window of message counters from that sender to determine whether it is a duplicate. The Message Layer SHALL discard duplicate messages before they reach the application layer.

4.6.5.1. Message Reception State

The state maintained by a Node about the messages it has received from a particular peer is referred to as *message reception state*. Nodes use this state information to determine whether a newly arrived message is a duplicate of a previous message. Message reception state is maintained on a per-peer or per-session basis, depending on the type of message encryption being used.

At a conceptual level, message reception state consists of a set of integers corresponding to the counters of all the messages that have been received from a particular peer. To limit the amount of memory required to store this state, Nodes employ a lossy compression scheme that takes advantage of the fact that message counters are generated sequentially by the sender. The scheme allows for a limited amount of out-of-order message arrivals due to network effects without inducing false detection of duplicates.

In the compressed form, message reception state is structured as a pair of values: a integer representing the largest valid, or maximum message counter received from the peer (**max_message_counter**), and a bitmap of size **MSG_COUNTER_WINDOW_SIZE** indicating which messages immediately prior to the max message have been received. The offset into the bitmap equates to the difference between the corresponding message counter and the max message counter, i.e. the first bit in the bitmap indicates whether the message with the counter of **max_message_counter - 1** has been received, the second indicates whether message **max_message_counter - 2** has been received, and so on. A message counter is within the range of the bitmap, also known as the message counter window, when the counter value is between $[(\text{max_message_counter} - \text{MSG_COUNTER_WINDOW_SIZE}) \text{ to } (\text{max_message_counter} - 1) \bmod 2^{32}]$. As messages arrive, the message reception state is queried to determine if an arriving message is new or duplicate. If a message is new, the state is then updated to reflect the arrival of the message. When a message arrives with a message counter that is logically greater than the current maximum message counter for that peer, the maximum message counter value for the peer is updated and the bitmap shifted accordingly.